Take Test: Recap Assignment

Test Information

Description **10% of final module grade.**

Instructions Answer ALL Questions.
Statistical Tables are added here for your convenience.
Standard_Normal_Tables.pdf
F_distribution_tables.pdf
t_distribution_tables.pdf

- Feedback is not provided immediately on submission. Submissions are marked manually not automated. Feedback will be provided when all submissions have been graded.
- While working on your submission you can click the SAVE button and return to your workings at a later time to continue (as long as it is before the due date).
- To complete your submission, i.e. submit your workings for grading, click the SUBMIT button.
- You should need to make only 1 submission. However, a maximum of 2 attempts is allowed, the 2nd attempt is a back up in case you inadvertently click submit before you were finished.

Multiple Attempts This Test allows 2 attempts. This is attempt number 2.

Force Completion This Test can be saved and resumed later.

Your answers are saved automatically.

QUESTION 1 **1 points** Save Answer

In a study into typical monthly mobile revenue per user (RPU), random samples of mobile customers were selected, 10 prepay customers and 7 billpay customers.

The usage in euro observed for those individual customers last month shows a right-skewed distribution.

Whichof following tests would you apply:

- Signs test for a median
- Wilcoxon rank sum test
- Pearson correlation test
- 2-sample t-test for comparing two population means
- 1-sample t-test for a mean
- Spearman Rank sum test

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

Save

The true correlation coefficient between two variables X and Y for a population of individuals is known to have value ρ = 0. This implies that X and Y are unrelated." Is this statement true or false?

- True
- False

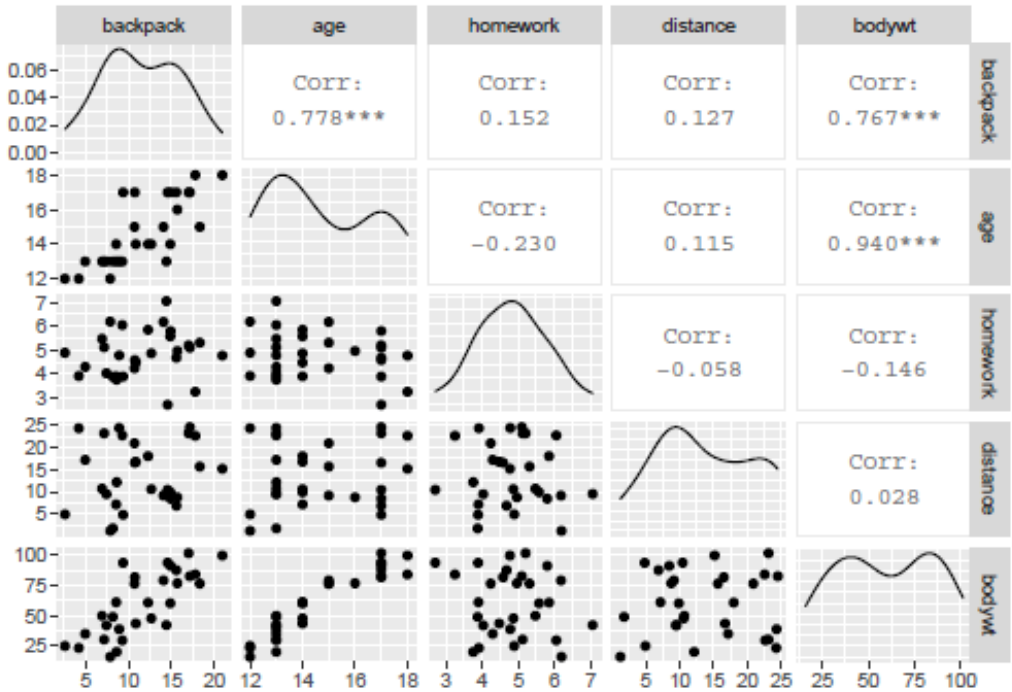
QUESTION 3

2 points Save Answer

Researchers running the School Years Welfare Study were interested in explaining the variability in weights (kgs) of backpacks carried by children, variable backpack. The following candidate predictor variables were recorded for a sample of 30 children:

- age : the age of the child in years
- bodywt : bodyweight in kgs.
- homework : number of hours homework per week
- distance : distance between home and school in kilometres.

Descriptive statistics and scatterplots for the quantitative variables are provided below.



The sample Pearson's correlation coefficient between backpack and age is reported to be r = 0.778, **interpret** the meaning of this value.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 4

6 points Save Answer

Carry out Pearson's correlation test to determine whether there is evidence of a significant increasing relationship between the variables backpack and age in the population. Include clear
statements of the hypotheses, the rejection region and conclusion.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 5

1 points Save Answer

Which of the following is a non-parametric alternative test to the Pearson's correlation test?

- Spearman's Rank test
- Signs test
- Wilcoxon Rank sum test
- 1-sample z-test
- 2-sample t-test
- ANOVA test

QUESTION 6

2 points Save Answer

A multiple linear regression model, using the three candidate predictors is proposed:
$\text{backpack}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{homework}_i + \beta_3 \text{distance}_i + \varepsilon_i$

What does the error term, ε_i, represent?

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 7

5 points Save Answer

When carrying out an Analysis of Variance for any regression analysis, **explain in words** what the terms SSTotal, SSReg and SSRes represent.

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QUESTION 8

4 points Save Answer

The model, $\text{backpack}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{homework}_i + \beta_3 \text{distance}_i + \varepsilon_i$, is fitted to the observations in the sample of 30 individuals and the results of the fit are provided in the partial printout below.

```
Coefficients:
               Estimate  Std. Error   t value   Pr(>|t|)
(Intercept)    -26.2531    4.8922    -5.3663    0.0000
age              2.0009    0.2503    7.9929    0.0000
homework         1.6775    0.5071      -          -
distance         0.0318    0.0666    0.4775    0.6370
---
Residual standard error:  -  on  -  degrees of freedom
Multiple R-squared:   -   ,    Adjusted R-squared:  -
F-statistic: 22.6  on  -  and  -  DF,  p-value: 0.0000002054334
```

Analysis of variance for the model fitted to the observed data gives the values SS_{Total} = 611.18 and SS_{Reg} = 441.78.

i) The coefficient of determination R^2 is missing from the output. Find this value.

ii) The value of the residual standard error, s, is missing from the output. Find this value.

QUESTION 9

2 points Save Answer

Interpret the meaning of the value of the coefficient of determination, R^2, in this application.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 10

3 points Save Answer

Give an interpretation for the fitted coefficient for each of the three predictor variables.

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QUESTION 11

6 points Save Answer

The test statistic for the ANOVA F-test is printed as 22.6. Show how this value is calculated.

Write down the hypotheses being tested.

Referencing the p-value provided, what is the conlcusion for this test in terms of the variables in the application.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 12

5 points Save Answer

Complete the missing t-test in the output, i.e. test whether homework is a good predictor of backpack at the $\alpha = 0.05$ level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 13

3 points Save Answer

What can you conclude as a result of the t-tests in this output? Choose one or more from the following:

- [ ] The t-test for $H_0 : \beta_2 = 0$ versus $H_a: \beta_2 \neq 0$ suggests there is evidence that homework is a good predictor of backpack weight.
- [ ] The t-test for $H_0 : \beta_2 = 0$ versus $H_a: \beta_2 \neq 0$ suggests there is no evidence that homework is a good predictor of backpack weight.
- [ ] The t-test for $H_0 : \beta_3 = 0$ versus $H_a: \beta_3 \neq 0$ suggests there is no evidence that distance is a good predictor of backpack weight.
- [ ] The t-test for $H_0 : \beta_3 = 0$ versus $H_a: \beta_3 \neq 0$ suggests there is evidence that distance is a good predictor of backpack weight.
- [ ] The t-test for $H_0 : \beta_3 = 0$ versus $H_a: \beta_3 \neq 0$ suggests there is evidence that distance is not a good predictor of backpack weight.
- [ ] The t-test for $H_0 : \beta_1 = 0$ versus $H_a: \beta_1 \neq 0$ suggests there is evidence that age is a good predictor of backpack weight.
- [ ] The t-test for $H_0 : \beta_1 = 0$ versus $H_a: \beta_1 \neq 0$ suggests there is no evidence that age is a good predictor of backpack weight.

QUESTION 14

2 points Save Answer

The output below provides a prediction interval and a co
response when age = 14, homework = 5 and distance = 11.

Prediction interval for the response:

```
       fit      lwr      upr
1 10.49695 5.142898 15.85099
```

Confidence interval for the response:

```
       fit      lwr      upr
1 10.49695 9.430513 11.56338
```

Explain, referencing the application, the difference as to when a prediction interval and a confidence interval applies.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 15

2 points Save Answer

In one sentence, explain the term "multicollinearity".

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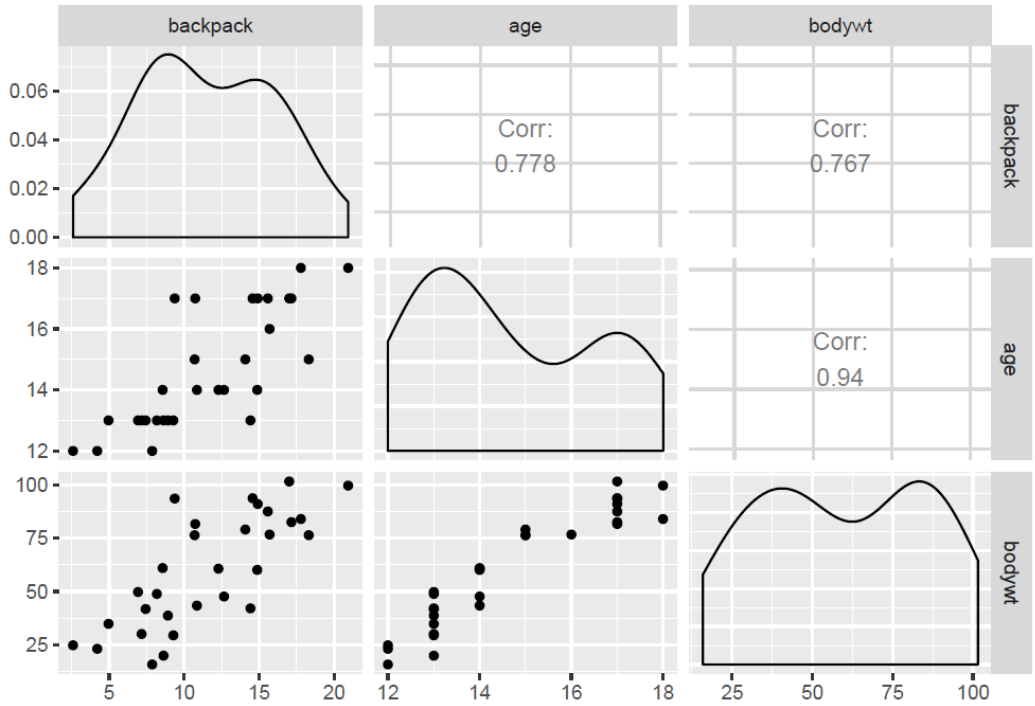
QUESTION 16

10 points Save Answer

The plot and output below is produced to assess a model where age and bodywt are used to predict backpack weight,

$backpack_i = \beta_0 + \beta_1 age_i + \beta_2 bodywt_i + \varepsilon_i$.




```
Call:
lm(formula = backpack ~ age + bodywt, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max
-6.6914 -1.9923  0.4405  1.6110  5.6846

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.42285    8.70075  -0.968    0.342
age          1.14748    0.82034   1.399    0.173
bodywt       0.05324    0.06059   0.879    0.387

Residual standard error: 2.949 on 27 degrees of freedom
Multiple R-squared:  0.6157,    Adjusted R-squared:  0.5873
F-statistic: 21.63 on 2 and 27 DF,  p-value: 0.00000247
```

Discuss the predictive power of the two variables, age and bodywt, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 17

2 points Save Answer

In one sentence, explain the meaning of the phrase "a parsimonious model" in variable selection.

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QUESTION 18

8 points Save Answer

A best-subsets routine was carried out on 4 candidate predictors, age, homework, distance and bodywt.

The best fitting model for each number of predictors is selected and summarised below.

model	predictors	p	s	Rsq	adjRsq	Cp	AIC	BIC
1	age	1	2.937	0.605	0.591	9.944	153.716	157.920
2	age homework	2	2.516	0.720	0.700	1.426	145.330	150.935
3	age homework distance	3	2.553	0.723	0.691	3.204	147.068	154.074
4	age homework distance bodywt	4	2.592	0.725	0.681	5.000	148.824	157.231

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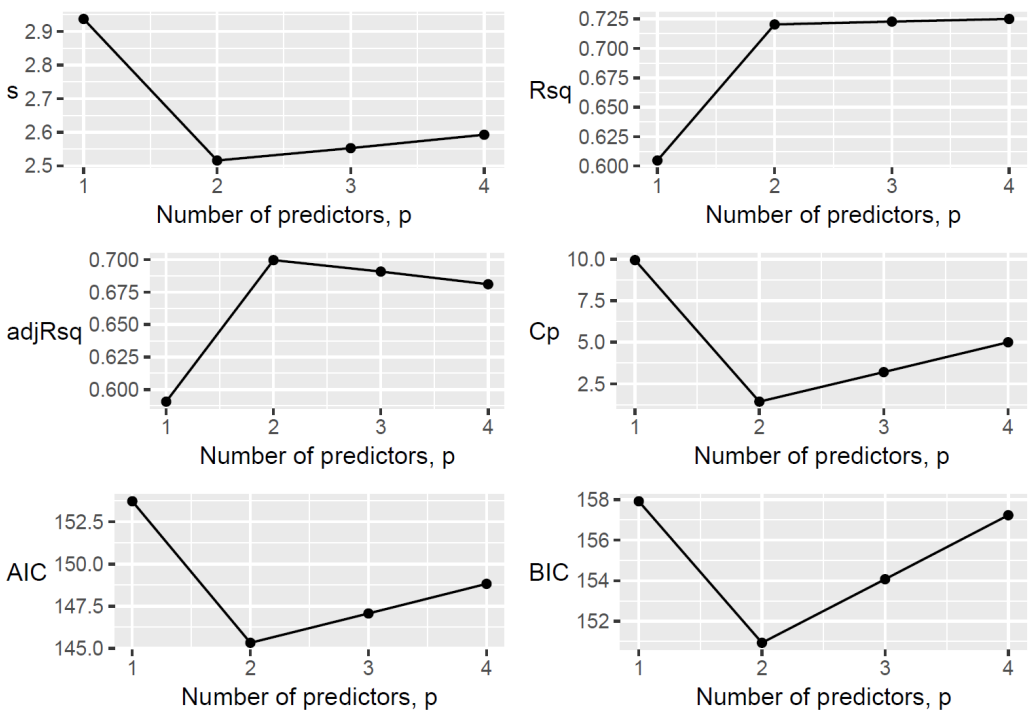

Using the output to support your answer, write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed

backpack. Justify your choice of model by explaining the role of s, R^2, adjusted R^2, and Mallow's Cp in choosing the best model.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 19

4 points Save Answer

When fitting a linear regression model to data what assumptions are required?

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QUESTION 20

3 points Save Answer

The percentage load of weight of the backpack, PercLoad, is calculated as the backpack weight relative to the individual's body mass.
The following output outlines the fit for the predictor age on the response PercLoad,

$PercLoad_i = \beta_0 + \beta_1 age_i + \varepsilon_i$

```
Call:
lm(formula = PercLoad ~ age, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.7928 -1.3114 -0.6993  1.1469  9.1218 

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)    
(Intercept)  23.2933     4.2454   5.487 0.00000735 ***
age          -0.9826     0.2876  -3.416    0.00196 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.03 on 28 degrees of freedom
Multiple R-squared:  0.2942,	Adjusted R-squared:  0.269 
F-statistic: 11.67 on 1 and 28 DF,  p-value: 0.001959
```

i) Use the estimated regression equation for this model to find the predicted value of PercLoad for a child aged 12 years old.

ii) The observed PercLoad for this child was 20.62. Calculate the fitted residual.

iii) Find the value of the standardised residual for this observation.

QUESTION 21

2 points Save Answer

Explain what assumption the standardised residuals are used to validate. What would you say about the observation in the previous question?

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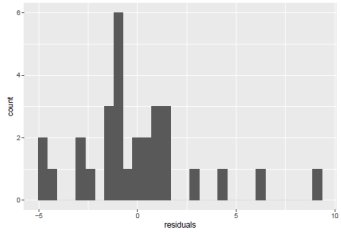
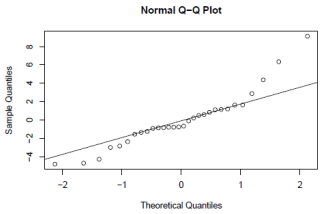
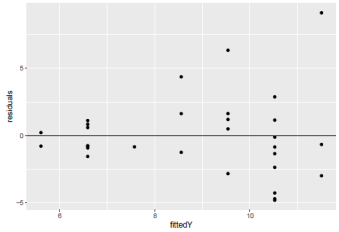
QUESTION 22

4 points Save Answer

Diagnositc Plots for the fitted regression model for PercLoad are produced:







Explain how the diagnostic plots are used to assess validity of the assumptions. Comment on the outputs of these diagnostic plots.

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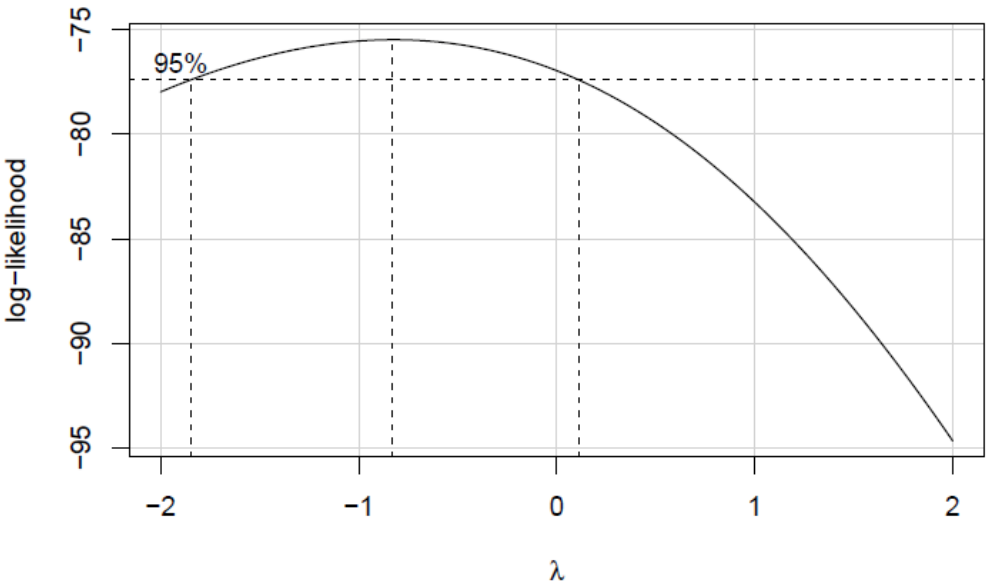
QUESTION 23

4 points Save Answer

What does the following box-cox transformation output indicate?

```
bcPower Transformation to Normality
   Est Power Rounded Pwr Wald Lwr bnd Wald Upr Bnd
Y1   -0.8236           0       -1.798       0.1507

Likelihood ratio tests about transformation parameters
                             LRT df         pval
LR test, lambda = (0)  2.959074  1 0.0853968227
LR test, lambda = (1) 15.545215  1 0.0000805553
```



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QUESTION 24

4 points Save Answer

A simple linear regression using age as the predictor was fitted to the transformed response variable ln(PercLoad) ,

$\ln(\text{PercLoad})_i = \beta_0 + \beta_1 \text{age}_i + \varepsilon_i$

giving the output below.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.7062	0.4123	8.9886	<0.0001
age	-0.1083	0.0279	-3.8752	0.0006

Using this model **predict** the percentage load of weight of the backpack of an individual aged 12 years.

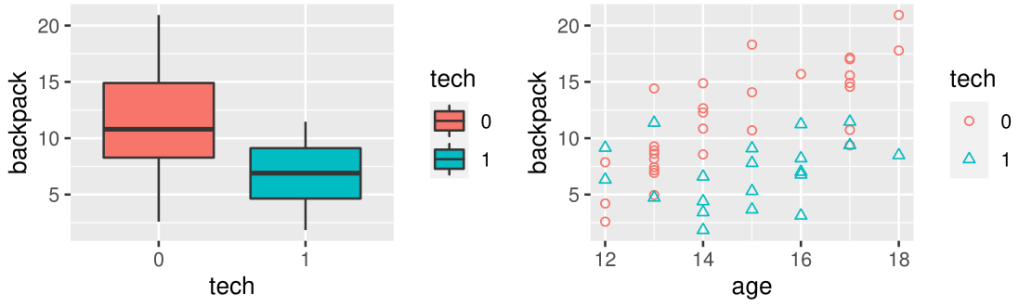
QUESTION 25

2 points Save Answer

Researchers wished to assess the effect on backpack weight when children use technology in their school day, e.g. iPad/tablet. Backpack weight was observed for children who do use technology and for children who don't use technology in

the school day, the variable tech is stored as an indicator variable defined as

- $tech_i=0$, if child i uses no technology and
- $tech_i=1$, if child i uses technology.



A model using the indicator variable tech, i.e. $backpack_i = \beta_0 + \beta_1 tech_i + \varepsilon_i$, is fitted to the data giving the output below.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	11.5554	0.7303	15.8233	<0.0001
tech	-4.5821	1.1547	-3.9683	0.0002

Using this fitted model provide the estimate of backpack weight for:

- a child that doesn't use technology.
- a child that does use technology.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 26

3 points Save Answer

Interpret the meaning of the estimated coefficient of tech and comment on the significance of the effect. Ensure you state the hypotheses being tested when referring to the results of the output.

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QUESTION 27

1 points Save Answer

This analysis is equivalent to which of the following two population comparison tests:

- [ ] Wilcoxon rank sum test for difference in population medians
- [ ] Independent sample t-test for difference in population means with unequal variance
- [ ] Independent sample t-test for difference in population means with equal variance
- [ ] Paired t-test for difference in population means

QUESTION 28

3 points Save Answer

A plot of the observed relationship of backpack weight and the age of the child was also provided, with an indication to those who did and didn't use tech.

Age is added to the model as a covariate of predictor tech, in the model

$backpack_i = \beta_0 + \beta_1 age_i + \beta_2 tech_i + \varepsilon_i$

and fitted to the data giving the following output.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-8.6823	3.6202	-2.3983	0.0205
age	1.3830	0.2443	5.6606	<0.0001
tech	-4.9509	0.9022	-5.4878	<0.0001

Using this fitted model provide the estimated relationship between backpack and age for:

- a child that doesn't use technology.
- a child that does use technology.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 29

4 points Save Answer

Interpret the meaning of the estimated coefficients of age and tech and comment on the significance of the effects. Ensure you state the hypotheses being tested when referring to the results of the output.

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QUESTION 30

2 points Save Answer

What assumption is being made in this model about the relationship between backpack weight and age when tech is used or not used?

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QUESTION 31

4 points Save Answer

An interaction term of age and tech, defined as age:tech$_i$=(age$_i$) x (tech$_i$), is added to the model

backpack$_i$ = $\beta_0 + \beta_1$age$_i$ + $\beta_2$tech$_i$ + β_3 age:tech$_i$ +ε_i

and fitted to the data giving the following output.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-15.1509	4.0718	-3.7210	0.0005
age	1.8250	0.2759	6.6153	<0.0001
tech	15.6276	7.2938	2.1426	0.0375
age:tech	-1.3890	0.4890	-2.8403	0.0067

Using this fitted model provide the estimated relationship between backpack weight and age for:

- a child that doesn't use technology.
- a child that does use technology.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 32

5 points Save Answer

Interpret the meaning of the fitted coefficient of age : tech in this model. Determine if the term is significant, using the result of the individual t-test in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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